Gear Train & Power Transmission Calculator

Engineering Notebook

Week 1

Date: September 2026

Project Goal

I wanted to better understand how mechanical systems transfer motion and power through gears.

Before starting this project, I knew that gears could change speed and torque, but I did not fully understand how those changes could be calculated or applied in real engineering systems.

What I Worked On

- Researched the basic principles of gear systems.
- Learned about gear ratios and how they affect speed and torque.
- Studied rotational speed (RPM) and torque relationships.
- Created the GitHub repository and project structure.
- Designed the project visual and README layout.

What I Learned

I learned that gears are not only used to transfer motion but also to modify the performance of a mechanical system.

A simple change in gear ratio can completely change the balance between speed and torque.

Challenges

At first, I found it difficult to understand why increasing torque normally decreases rotational speed.

Results

- Research phase completed.
- Project structure completed.
- Basic gear ratio formulas identified.

Next Steps

- Build the first version of the calculator.
- Calculate gear ratio.
- Calculate output RPM and output torque.

Week 2

Date: September 2026

What I Worked On

- Developed the first working version of the calculator in Python.
- Added user inputs for:
    - Driver Gear Teeth
    - Driven Gear Teeth
    - Input RPM
    - Input Torque
- Calculated gear ratio.
- Calculated output RPM.
- Calculated output torque.
- Improved README documentation.
- Added project visuals and explanations.

What I Learned

Working on this project helped me understand how engineers use mathematical relationships to analyze and design mechanical systems.

I also learned that mechanical systems often involve trade-offs. Increasing torque usually means reducing speed, while increasing speed usually reduces torque.

Results

The calculator successfully evaluates the effect of gear ratios on system performance.

Example

Driver Gear Teeth = 12

Driven Gear Teeth = 48

Input RPM = 1200

Input Torque = 10 Nm

Output

Gear Ratio = 4:1

Output RPM = 300

Output Torque = 40 Nm

Challenges

One of the biggest challenges was understanding how the same gear system could improve one performance parameter while reducing another.

Key Lessons Learned

- Gear ratios directly affect speed and torque.
- Mechanical systems are built around trade-offs.
- Mathematics helps engineers predict system behavior.
- Programming can be used to model mechanical processes.
- Small design changes can produce large performance differences.

Final Reflection

Before starting this project, I mostly thought of gears as simple machine components.

As I learned more about gear ratios, RPM, torque, and power transmission, I realized how important gears are in engineering design. What appears to be a simple mechanism can completely change how a machine behaves.

Building this calculator helped me better understand how engineers approach mechanical problems and use mathematics to optimize performance.

The most rewarding part was seeing how a few engineering equations could explain the behavior of real-world mechanical systems.

It reminded me that engineering is not only about building machines, but also about understanding how systems work and how simple ideas can be used to solve complex problems. **Engineering is the art of transforming motion.**